Sanam Rai
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GitHub | LinkedIn

Professional Summary

Bachelor of Information Technology (BIT) student with strong expertise in backend development and Software Quality Assurance. Experienced in designing scalable APIs, implementing authentication systems, and building modular full-stack applications using the MERN stack. Skilled in testing methodologies, automation, and debugging workflows. Passionate about backend architecture, system design, and developing secure, production-ready systems.

Technical Skills

Languages: Python, JavaScript (ES6+), HTML5, CSS3
Backend: Node.js, Express.js, Flask
Frontend: React.js, Tailwind CSS
Databases: MongoDB, MySQL
Testing & QA: Selenium (Python), Playwright (Python), Manual Testing, Functional & Regression Testing, Bug Reporting
Tools: Git, GitHub, Postman, VS Code, Pytest

Professional Experience

Quality Assurance Intern
Danfe Solution Pvt. Ltd., Kathmandu, Nepal
Dec 2025 – Mar 2026

- Conducted manual functional testing across multiple internal systems, identifying and documenting defects with clear reproduction steps.
- Performed regression and validation testing to ensure application stability after updates.
- Collaborated with developers to debug issues and improve product quality.
- Created technical documentation to streamline workflows and internal processes.
- Assisted in client onboarding and product training, improving user adoption and understanding.

Education

Bachelor in Information Technology (BIT)
Tribhuvan University – Bhaktapur Multiple Campus
Expected Graduation: mid 2026

Relevant Coursework:
Data Structures & Algorithms, DBMS, Software Engineering, Operating Systems, Computer Networks

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Projects

Portfolio System (Feature Toggle Architecture) | GitHub Link

- Built a backend-driven portfolio system with **dynamic feature toggles** (auth, caching, rate limiting, pagination, logging).
- Implemented **JWT-based authentication middleware** with toggle-based enable/disable control.
- Designed **in-memory caching system** to reduce database load and improve performance.
- Developed **custom pagination logic** supporting query-based page navigation.
- Integrated **rate limiting middleware** to simulate production-level API protection.
- Structured backend using clean modular architecture (controllers, middleware, config layers).

YakTalk – Secure Real-Time Chat Application | GitHub Link

- Developed a real-time chat application using MERN stack and Socket.io.
- Implemented JWT authentication and secured passwords using bcrypt hashing.
- Added real-time online presence tracking and private messaging system.

Phone Book Application | GitHub Link

- Built CLI-based contact management system with full CRUD operations.
- Used Regex for input validation and JSON for persistent storage.
- Improved reliability through unit testing with Pytest.

Certifications

- CS50x: Introduction to Computer Science – Harvard University (edX)
- CS50p: Introduction to Programming with Python – Harvard University (edX)